Unit: Unit 3: Functions
Topic: Function Transformations

Name: ________________
Date: ____________

Challenge (Habanero)

Q1. A recipe for making cookies is given by $y = 2x + 5$. If the recipe is scaled up to make twice as many cookies, the new equation is
(A) $y = 2x + 10$
(B) $y = 4x + 5$
(C) $y = 2x + 10$
(D) $y = x + 10$
(E) $y = 4x + 10$

Q2. The function $y = x^2$ is transformed into $y = (x-3)^2$. What is the horizontal shift?
(A) 3 units left
(B) 3 units right
(C) 2 units left
(D) 2 units right
(E) 1 unit left

Q3. A builder uses the function $y = 2x$ to design a roof. If the design is reflected over the x-axis, the new equation is
(A) $y = -2x$
(B) $y = 2x$
(C) $y = -x$
(D) $y = x$
(E) $y = 2x + 1$

Q4. A chef uses the function $y = x^2$ to model the shape of a cake. If the cake is compressed horizontally by a factor of 2, the new equation is
(A) $y = (2x)^2$
(B) $y = (x/2)^2$
(C) $y = x^2$
(D) $y = 2x^2$
(E) $y = x^2/2$

Q5. The function $y = x^2 + 1$ is transformed into $y = x^2 - 1$. What is the vertical shift?
(A) 2 units down
(B) 2 units up
(C) 1 unit down
(D) 1 unit up
(E) 3 units down

Q6. A contractor uses the function $y = 3x$ to model the height of a building. If the building is reflected over the y-axis, the new equation is
(A) $y = -3x$
(B) $y = 3x$
(C) $y = -x$
(D) $y = x$
(E) $y = 3x + 1$

Q7. The function $y = x^2$ is transformed into $y = (x + 2)^2 - 3$. What are the horizontal and vertical shifts?
(A) 2 units left, 3 units down
(B) 2 units right, 3 units down
(C) 2 units left, 3 units up
(D) 2 units right, 3 units up
(E) 1 unit left, 2 units down

Q8. A recipe for making bread is given by $y = x + 2$. If the recipe is scaled up to make 1.5 times as much bread, the new equation is
(A) $y = 1.5x + 2$
(B) $y = x + 3$
(C) $y = 1.5x + 3$
(D) $y = x + 1.5$
(E) $y = 1.5x + 1$

Open Ended Questions (FRQ)

Q9. Describe the transformations applied to the function $y = x^2$ to obtain the function $y = (x-2)^2+1$. Provide a clear explanation of the horizontal and vertical shifts, and any other transformations applied.

Q10. Graph the functions $y = x^2$ and $y = (x + 1)^2 - 2$ on the same coordinate plane. Compare the graphs and describe any similarities or differences in terms of transformations applied.

Chef's Tips

- Check for proper labeling of axes and graph titles
- Look for clear explanations of transformations applied
- Verify accuracy of graphed functions